

Newman-Penrose Formalism Calculations

Stephani et al., Equation (21.11)

Metric

$$ds^2 = \left(\frac{X(p) - Y(q)}{p^2q^4 - 2p^3q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\tau \otimes d\tau + \left(\frac{q^2X(p) + p^2Y(q)}{p^2q^4 - 2p^3q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\tau \otimes d\sigma + \left(\frac{q^2X(p) + p^2Y(q)}{p^2q^4 - 2p^3q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\sigma \otimes d\tau + \left(\frac{q^4X(p) - p^4Y(q)}{p^2q^4 - 2p^3q - 2pq^3 + (p^4 + 1)q^2 + p^2} \right) d\sigma \otimes d\sigma + \left(\frac{p^2 + q^2}{p^2q^2X(p) - 2pqX(p) + X(p)} \right) dp \otimes dp + \left(\frac{p^2 + q^2}{(p^2q^2 - 2pq + 1)Y(q)} \right) dq \otimes dq$$

Coordinates

$$(x^\mu) = (\tau, \sigma, p, q)$$

Null Tetrad

Covariant components

$$l = \left(\frac{\sqrt{\frac{1}{2}}}{\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(\frac{\sqrt{\frac{1}{2}}q^4X(p)}{\left(q^2X(p) + p^2Y(q) + \sqrt{X(p)Y(q)}p^2 + \sqrt{X(p)Y(q)}q^2 \right)\sqrt{p^2 + q^2}(pq - 1)} - \frac{\sqrt{\frac{1}{2}}p^4Y(q)}{\left(q^2X(p) + p^2Y(q) + \sqrt{X(p)Y(q)}p^2 + \sqrt{X(p)Y(q)}q^2 \right)\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

$$n = \left(-\frac{\sqrt{\frac{1}{2}}X(p)}{\sqrt{p^2 + q^2}(pq - 1)} + \frac{\sqrt{\frac{1}{2}}Y(q)}{\sqrt{p^2 + q^2}(pq - 1)} \right) d\tau + \left(-\frac{\sqrt{\frac{1}{2}}q^2X(p)}{\sqrt{p^2 + q^2}(pq - 1)} - \frac{\sqrt{\frac{1}{2}}p^2Y(q)}{\sqrt{p^2 + q^2}(pq - 1)} - \frac{\sqrt{\frac{1}{2}}\sqrt{X(p)Y(q)}p^2}{\sqrt{p^2 + q^2}(pq - 1)} - \frac{\sqrt{\frac{1}{2}}\sqrt{X(p)Y(q)}q^2}{\sqrt{p^2 + q^2}(pq - 1)} \right) d\sigma$$

$$m = \left(-\frac{\sqrt{\frac{1}{2}}\sqrt{\frac{p^2}{X(p)} + \frac{q^2}{X(p)}}}{pq - 1} \right) dp + \left(-\frac{i\sqrt{\frac{1}{2}}\sqrt{\frac{p^2}{Y(q)} + \frac{q^2}{Y(q)}}}{pq - 1} \right) dq$$

$$\overline{m} = \left(-\frac{\sqrt{\frac{1}{2}}\sqrt{\frac{p^2}{X(p)} + \frac{q^2}{X(p)}}}{pq - 1} \right) dp + \left(\frac{i\sqrt{\frac{1}{2}}\sqrt{\frac{p^2}{Y(q)} + \frac{q^2}{Y(q)}}}{pq - 1} \right) dq$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}\sqrt{p^2 + q^2}pq^3\sqrt{X(p)}}{2\left(p^2X(p)\sqrt{Y(q)} + q^2X(p)\sqrt{Y(q)} + p^2\sqrt{X(p)}Y(q) + q^2\sqrt{X(p)}Y(q)\right)} + \frac{\sqrt{2}\sqrt{p^2 + q^2}p^3q\sqrt{Y(q)}}{2\left(p^2X(p)\sqrt{Y(q)} + q^2X(p)\sqrt{Y(q)} + p^2\sqrt{X(p)}Y(q) + q^2\sqrt{X(p)}Y(q)\right)} \right. \\ \left. + \frac{\sqrt{2}\sqrt{p^2 + q^2}q^2\sqrt{X(p)}}{2\left(p^2X(p)\sqrt{Y(q)} + q^2X(p)\sqrt{Y(q)} + p^2\sqrt{X(p)}Y(q) + q^2\sqrt{X(p)}Y(q)\right)} - \frac{\sqrt{2}\sqrt{p^2 + q^2}p^2\sqrt{Y(q)}}{2\left(p^2X(p)\sqrt{Y(q)} + q^2X(p)\sqrt{Y(q)} + p^2\sqrt{X(p)}Y(q) + q^2\sqrt{X(p)}Y(q)\right)} \right) d\tau \\ + \left(\frac{\sqrt{2}pq}{2\sqrt{p^2 + q^2}\sqrt{X(p)}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2 + q^2}\sqrt{X(p)}\sqrt{Y(q)}} \right) d\sigma$$

$$n = \left(-\frac{\sqrt{2}p^3q}{2\sqrt{p^2 + q^2}} - \frac{\sqrt{2}pq^3}{2\sqrt{p^2 + q^2}} - \frac{\sqrt{2}pq^3\sqrt{X(p)}}{2\sqrt{p^2 + q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}p^3q\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}\sqrt{X(p)}} + \frac{\sqrt{2}p^2}{2\sqrt{p^2 + q^2}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2 + q^2}} + \frac{\sqrt{2}q^2\sqrt{X(p)}}{2\sqrt{p^2 + q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}p^2\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}\sqrt{X(p)}} \right) d\tau \\ + \left(\frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2 + q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2 + q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2 + q^2}\sqrt{X(p)}} \right) d\sigma$$

$$m = \left(-\frac{\sqrt{p^2 + q^2}pq\sqrt{X(p)}}{\sqrt{2}p^2 + \sqrt{2}q^2} + \frac{\sqrt{p^2 + q^2}\sqrt{X(p)}}{\sqrt{2}p^2 + \sqrt{2}q^2} \right) dp + \left(-\frac{i\sqrt{p^2 + q^2}pq\sqrt{Y(q)}}{\sqrt{2}p^2 + \sqrt{2}q^2} + \frac{i\sqrt{p^2 + q^2}\sqrt{Y(q)}}{\sqrt{2}p^2 + \sqrt{2}q^2} \right) dq$$

$$\overline{m} = \left(-\frac{\sqrt{p^2 + q^2}pq\sqrt{X(p)}}{\sqrt{2}p^2 + \sqrt{2}q^2} + \frac{\sqrt{p^2 + q^2}\sqrt{X(p)}}{\sqrt{2}p^2 + \sqrt{2}q^2} \right) dp + \left(\frac{i\sqrt{p^2 + q^2}pq\sqrt{Y(q)}}{\sqrt{2}p^2 + \sqrt{2}q^2} - \frac{i\sqrt{p^2 + q^2}\sqrt{Y(q)}}{\sqrt{2}p^2 + \sqrt{2}q^2} \right) dq$$

Directional Derivatives

$$\begin{aligned}
D = & -\frac{\sqrt{2}\sqrt{p^2+q^2}pq^3\sqrt{X(p)}}{2\left(p^2X(p)\sqrt{Y(q)}+q^2X(p)\sqrt{Y(q)}+p^2\sqrt{X(p)}Y(q)+q^2\sqrt{X(p)}Y(q)\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^3q\sqrt{Y(q)}}{2\left(p^2X(p)\sqrt{Y(q)}+q^2X(p)\sqrt{Y(q)}+p^2\sqrt{X(p)}Y(q)+q^2\sqrt{X(p)}Y(q)\right)} \\
& + \frac{\sqrt{2}\sqrt{p^2+q^2}q^2\sqrt{X(p)}}{2\left(p^2X(p)\sqrt{Y(q)}+q^2X(p)\sqrt{Y(q)}+p^2\sqrt{X(p)}Y(q)+q^2\sqrt{X(p)}Y(q)\right)} \\
& - \frac{\sqrt{2}\sqrt{p^2+q^2}p^2\sqrt{Y(q)}}{2\left(p^2X(p)\sqrt{Y(q)}+q^2X(p)\sqrt{Y(q)}+p^2\sqrt{X(p)}Y(q)+q^2\sqrt{X(p)}Y(q)\right)}\partial_\tau + \frac{\sqrt{2}pq}{2\sqrt{p^2+q^2}\sqrt{X(p)}\sqrt{Y(q)}} - \frac{\sqrt{2}}{2\sqrt{p^2+q^2}\sqrt{X(p)}\sqrt{Y(q)}}\partial_\sigma
\end{aligned}$$

$$\begin{aligned}
\Delta = & -\frac{\sqrt{2}p^3q}{2\sqrt{p^2+q^2}} - \frac{\sqrt{2}pq^3}{2\sqrt{p^2+q^2}} - \frac{\sqrt{2}pq^3\sqrt{X(p)}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}p^3q\sqrt{Y(q)}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{\sqrt{2}p^2}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}q^2}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}q^2\sqrt{X(p)}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} \\
& + \frac{\sqrt{2}p^2\sqrt{Y(q)}}{2\sqrt{p^2+q^2}\sqrt{X(p)}}\partial_\tau + \frac{\sqrt{2}pq\sqrt{X(p)}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}pq\sqrt{Y(q)}}{2\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{\sqrt{2}\sqrt{X(p)}}{2\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}\sqrt{Y(q)}}{2\sqrt{p^2+q^2}\sqrt{X(p)}}\partial_\sigma
\end{aligned}$$

$$\delta = -\frac{\sqrt{p^2+q^2}pq\sqrt{X(p)}}{\sqrt{2}p^2+\sqrt{2}q^2} + \frac{\sqrt{p^2+q^2}\sqrt{X(p)}}{\sqrt{2}p^2+\sqrt{2}q^2}\partial_p - \frac{i\sqrt{p^2+q^2}pq\sqrt{Y(q)}}{\sqrt{2}p^2+\sqrt{2}q^2} + \frac{i\sqrt{p^2+q^2}\sqrt{Y(q)}}{\sqrt{2}p^2+\sqrt{2}q^2}\partial_q$$

$$\bar{\delta} = -\frac{\sqrt{p^2+q^2}pq\sqrt{X(p)}}{\sqrt{2}p^2+\sqrt{2}q^2} + \frac{\sqrt{p^2+q^2}\sqrt{X(p)}}{\sqrt{2}p^2+\sqrt{2}q^2}\partial_p + \frac{i\sqrt{p^2+q^2}pq\sqrt{Y(q)}}{\sqrt{2}p^2+\sqrt{2}q^2} - \frac{i\sqrt{p^2+q^2}\sqrt{Y(q)}}{\sqrt{2}p^2+\sqrt{2}q^2}\partial_q$$

Spin Coefficients

$$\rho = 0$$

$$\sigma = 0$$

$$\begin{aligned}
\alpha = & \frac{\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^2X(p)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q^3X(p)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2q\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^2\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q^3\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3Y(q)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2qY(q)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q^2\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \dots
\end{aligned}$$

$$\begin{aligned}
\beta = & \frac{\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3q\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^3\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^2X(p)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}q^3X(p)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2q\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}pq^2\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}q^3\sqrt{X(p)}\sqrt{Y(q)}}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& + \frac{i\sqrt{2}\sqrt{p^2+q^2}p^3Y(q)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} + \frac{\sqrt{2}\sqrt{p^2+q^2}p^2qY(q)}{4\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \frac{\sqrt{2}\sqrt{p^2+q^2}q^2\frac{\partial}{\partial p}X(p)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} \\
& - \frac{i\sqrt{2}\sqrt{p^2+q^2}p^2\frac{\partial}{\partial q}Y(q)}{8\left(p^4\sqrt{X(p)}+2p^2q^2\sqrt{X(p)}+q^4\sqrt{X(p)}+p^4\sqrt{Y(q)}+2p^2q^2\sqrt{Y(q)}+q^4\sqrt{Y(q)}\right)} - \dots
\end{aligned}$$

$$\tau = \frac{\sqrt{2}pq\frac{\partial}{\partial p}X(p)}{8\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}pq\frac{\partial}{\partial q}Y(q)}{8\sqrt{p^2+q^2}\sqrt{Y(q)}} - \frac{\sqrt{2}q\sqrt{X(p)}}{2\sqrt{p^2+q^2}} - \frac{i\sqrt{2}p\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} - \frac{\sqrt{2}\frac{\partial}{\partial p}X(p)}{8\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}\frac{\partial}{\partial q}Y(q)}{8\sqrt{p^2+q^2}\sqrt{Y(q)}}$$

$$\pi = -\frac{\sqrt{2}pq\frac{\partial}{\partial p}X(p)}{8\sqrt{p^2+q^2}\sqrt{X(p)}} + \frac{i\sqrt{2}pq\frac{\partial}{\partial q}Y(q)}{8\sqrt{p^2+q^2}\sqrt{Y(q)}} + \frac{\sqrt{2}q\sqrt{X(p)}}{2\sqrt{p^2+q^2}} - \frac{i\sqrt{2}p\sqrt{Y(q)}}{2\sqrt{p^2+q^2}} + \frac{\sqrt{2}\frac{\partial}{\partial p}X(p)}{8\sqrt{p^2+q^2}\sqrt{X(p)}} - \frac{i\sqrt{2}\frac{\partial}{\partial q}Y(q)}{8\sqrt{p^2+q^2}\sqrt{Y(q)}}$$

$$\epsilon = 0$$

$$\gamma = 0$$

Ricci Tensor Components

$$\begin{aligned}
\Phi_{00} = & - \frac{p^4 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{8 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& - \frac{p^2 q^4 \frac{\partial^2}{(\partial p)^2} X(p)}{8 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{p^4 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{8 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{p^2 q^4 \frac{\partial^2}{(\partial q)^2} Y(q)}{8 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{3 p^3 q^2 \frac{\partial}{\partial p} X(p)}{4 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{p q^4 \frac{\partial}{\partial p} X(p)}{4 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& - \frac{p^4 q \frac{\partial}{\partial q} Y(q)}{4 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& - \frac{3 p^2 q^3 \frac{\partial}{\partial q} Y(q)}{4 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& - \frac{3 p^2 q^2 X(p)}{2 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{3 p^2 q^2 Y(q)}{2 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} \\
& + \frac{p^3 q \frac{\partial^2}{(\partial p)^2} X(p)}{4 \left(p^4 X(p) + 2 p^2 q^2 X(p) + q^4 X(p) + 2 p^4 \sqrt{X(p)} \sqrt{Y(q)} + 4 p^2 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + p^4 Y(q) + 2 p^2 q^2 Y(q) + q^4 Y(q) \right)} + \dots
\end{aligned}$$

$$\Phi_{01} = 0$$

$$\Lambda = -\frac{p^2 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{24(p^2 + q^2)} - \frac{p^2 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^2 + q^2)} + \frac{pq^2 \frac{\partial}{\partial p} X(p)}{4(p^2 + q^2)} + \frac{p^2 q \frac{\partial}{\partial q} Y(q)}{4(p^2 + q^2)} - \frac{q^2 X(p)}{2(p^2 + q^2)} - \frac{p^2 Y(q)}{2(p^2 + q^2)} + \frac{pq \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^2 + q^2)} + \frac{pq \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^2 + q^2)} - \frac{q \frac{\partial}{\partial p} X(p)}{4(p^2 + q^2)} - \frac{p \frac{\partial}{\partial q} Y(q)}{4(p^2 + q^2)} - \frac{\frac{\partial^2}{(\partial p)^2} X(p)}{24(p^2 + q^2)} - \frac{\frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^2 + q^2)}$$

Weyl Tensor Components

$$\begin{aligned} \Psi_0 = & -\frac{p^6 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{8 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{p^4 q^4 \frac{\partial^2}{(\partial p)^2} X(p)}{4 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{p^2 q^6 \frac{\partial^2}{(\partial p)^2} X(p)}{8 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{p^6 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{8 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{p^4 q^4 \frac{\partial^2}{(\partial q)^2} Y(q)}{4 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{p^2 q^6 \frac{\partial^2}{(\partial q)^2} Y(q)}{8 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & +\frac{3 p^5 q^2 \frac{\partial}{\partial p} X(p)}{4 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} \\ & -\frac{3 i p^4 q^3 \frac{\partial}{\partial p} X(p)}{4 \left(p^6 X(p) + 3 p^4 q^2 X(p) + 3 p^2 q^4 X(p) + q^6 X(p) + 2 p^6 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^4 q^2 \sqrt{X(p)} \sqrt{Y(q)} + 6 p^2 q^4 \sqrt{X(p)} \sqrt{Y(q)} + 2 q^6 \sqrt{X(p)} \sqrt{Y(q)} + p^6 Y(q) + 3 p^4 q^2 Y(q) + 3 p^2 q^4 Y(q) + q^6 Y(q) \right)} + \dots \end{aligned}$$

$$\Psi_1 = 0$$

$$\begin{aligned}
\Psi_2 = & -\frac{p^6 q^2 \frac{\partial^2}{(\partial p)^2} X(p)}{24(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 \frac{\partial^2}{(\partial p)^2} X(p)}{24(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^6 q^2 \frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& - \frac{p^2 q^6 \frac{\partial^2}{(\partial q)^2} Y(q)}{24(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^5 q^3 \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{i p^4 q^3 \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^3 q^4 \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{i p^2 q^5 \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& + \frac{i p^5 q^2 \frac{\partial}{\partial q} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q^3 \frac{\partial}{\partial q} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{i p^3 q^4 \frac{\partial}{\partial q} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^2 q^5 \frac{\partial}{\partial q} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^2 X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{i p^3 q^3 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} \\
& + \frac{p^2 q^4 X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^4 q^2 Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{i p^3 q^3 Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{p^2 q^4 Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^5 q \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^3 q^3 \frac{\partial^2}{(\partial p)^2} X(p)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& + \frac{p q^5 \frac{\partial^2}{(\partial p)^2} X(p)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^5 q \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^3 q^3 \frac{\partial^2}{(\partial q)^2} Y(q)}{6(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p q^5 \frac{\partial^2}{(\partial q)^2} Y(q)}{12(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& + \frac{i p^3 q^2 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^3 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{i p q^4 \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{i p^4 q \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^3 q^2 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& - \frac{i p^2 q^3 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p q^4 \frac{\partial}{\partial q} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{p^3 q X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{2i p^2 q^2 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{p q^3 X(p)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} - \frac{p^3 q Y(q)}{p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6} + \dots
\end{aligned}$$

$$\Psi_3 = 0$$

$$\begin{aligned}
\Psi_4 = & -\frac{p^6 q^2 X(p) \frac{\partial^2}{(\partial p)^2} X(p)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 X(p) \frac{\partial^2}{(\partial p)^2} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 X(p) \frac{\partial^2}{(\partial p)^2} X(p)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^6 q^2 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial p)^2} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial p)^2} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial p)^2} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& - \frac{p^6 q^2 Y(q) \frac{\partial^2}{(\partial p)^2} X(p)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 Y(q) \frac{\partial^2}{(\partial p)^2} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 Y(q) \frac{\partial^2}{(\partial p)^2} X(p)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^6 q^2 X(p) \frac{\partial^2}{(\partial q)^2} Y(q)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 X(p) \frac{\partial^2}{(\partial q)^2} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 X(p) \frac{\partial^2}{(\partial q)^2} Y(q)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& - \frac{p^6 q^2 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial q)^2} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial q)^2} Y(q)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial^2}{(\partial q)^2} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^6 q^2 Y(q) \frac{\partial^2}{(\partial q)^2} Y(q)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^4 q^4 Y(q) \frac{\partial^2}{(\partial q)^2} Y(q)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{p^2 q^6 Y(q) \frac{\partial^2}{(\partial q)^2} Y(q)}{8(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& + \frac{3p^5 q^2 X(p) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{3i p^4 q^3 X(p) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{3p^3 q^4 X(p) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{3i p^2 q^5 X(p) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{3p^5 q^2 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{3i p^4 q^3 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} \\
& + \frac{3p^3 q^4 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{3i p^2 q^5 \sqrt{X(p)} \sqrt{Y(q)} \frac{\partial}{\partial p} X(p)}{2(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{3p^5 q^2 Y(q) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \frac{3i p^4 q^3 Y(q) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} + \frac{3p^3 q^4 Y(q) \frac{\partial}{\partial p} X(p)}{4(p^6 + 3p^4 q^2 + 3p^2 q^4 + q^6)} - \dots
\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"